

PMG Cybersecurity Profile

IT/SOC Team Overview

Piedmont Medical Centre maintains a dedicated and well-structured IT and Security Operations Center (SOC) team to ensure the confidentiality, integrity, and availability of its clinical and administrative systems. The IT department comprises 21 professionals distributed as follows:

- Network Engineers (5): Responsible for design, operation, and security of campus-wide networking, including VPN and wired/wireless infrastructure.
- System Administrators (4): Oversee maintenance and administration of Windows, Linux, and cloud-based systems supporting medical devices and applications.
- Database and Applications Specialists (3): Manage Electronic Medical Records (EMR), Electronic Health Records (EHR), and clinical application security.
- Help Desk (4): First-line support for all medical and administrative staff across multiple locations.
- IT Management (2): Information Technology leadership including CIO and IT Operations Manager oversee strategic initiatives.
- Specialists (3): Subject matter experts in telemedicine, radiology systems, and clinical research IT support.

The Cybersecurity team operates with the following key roles:

- Chief Information Security Officer (CISO): Provides overarching governance and compliance oversight.
- SOC Manager: Leads daily SOC operations and incident coordination.
- SOC Analysts (3): Deliver 24x7 monitoring, analysis, and response.
- Incident Response Lead: Orchestrates containment and remediation efforts.
- Governance, Risk, and Compliance (GRC) Specialist: Ensures continuous regulatory compliance and policy updates.

SOC functions include real-time monitoring of network traffic, endpoints, cloud platforms, and medical IoT devices; phishing incident triage; threat hunting; and staff security education.

HIPAA and Compliance Environment

Piedmont Medical Centre operates fully within the framework established by the Health Insurance Portability and Accountability Act (HIPAA), the Health Information Technology for Economic and Clinical Health Act (HITECH), and applicable state privacy laws. Compliance efforts focus on:

- Protecting Protected Health Information (PHI) through access controls, encryption, and robust logging.
- Mandatory annual HIPAA security and privacy training for all employees.
- Quarterly risk assessments and penetration tests to identify vulnerabilities.
- Enforcement of role-based access control policies across all clinical and administrative systems.
- Incident response policies aligned with HIPAA breach notification requirements.
- Regular internal and external audits verifying compliance.

Existing Monitoring Tools and Technologies

To safeguard its digital assets, Piedmont Medical Centre deploys an integrated suite of monitoring and security tools tailored to the healthcare sector, including:

- Security Information and Event Management (SIEM): Splunk Healthcare Edition aggregates logs from EMR/EHR systems, firewalls, endpoints, and cloud services, providing real-time alerting and compliance dashboards.
- Endpoint Detection & Response (EDR): SentinelOne deployed on all workstations, servers, and critical medical devices for active threat detection and containment.
- Perimeter Security: Next-generation firewalls with intrusion prevention systems (IPS), malware sandboxing, and VPN gateways with multi-factor authentication (MFA).

- **Threat Intelligence:** Integration with Health-ISAC, CISA advisories, and other commercial threat feed providers to stay current on healthcare-specific threats.
- **Vulnerability Management:** Regular Qualys scans identify and track remediation of known vulnerabilities.
- **Email Security:** Microsoft Defender for Office 365 provides filtering and anti-phishing protections tailored to medical communications.
- **Cloud Security Monitoring:** Azure Security Center offers continual assessment for patient portal and cloud-hosted applications.